# Samuel Crabbendam

crabbendam.net/sam | US Citizen | 407-973-5067 | linkedin.com/in/samuel-crabbendam-93ab33349/ | crabbendams@gmail.com

Portfolio

## Experience

**Payload Mechanics Responsible Engineer |** *Swamp Launch Rocketry, IREC Team*
Gainesville, FL | August **2024 –** Present

- Designing & integrating a rotating sensor module on 2027 Intercollegiate Rocket Engineering Competition (IREC) payload with Fusion360 CAD in coordination with electronics subteams, defining ± 0.01 in. part tolerances for airtight sealing.
- Selected Carbon Fiber Nylon from a filament material trade study for complex part geometries & 25 ksi tensile strength.
- Applied manufacturing principles to create a producible design while tracking revisions & releasing part drawings & BOMs.
- Designed the 2026 aluminum & PLA 4U CubeSat structure housing two experiments; producing 17 parts to ± 0.01 in.
- Designed 30 ± 5 °C thermoelectric thermal management system, protecting bio-research payloads during 6 g launch loads.
- Designed and machined the 2025 IREC aluminum and ABS chase camera deployment system over 3 design iterations.

**Project Engineering Intern |** *Garney Construction*
Tampa, FL | May **2026** – August **2026**

- Performed QC inspections & testing against contractor specifications on 4 structural assemblies weekly, alongside field crew.
- Verified GD&T on drawings supporting ± 2 in. design-to-production tolerances across 11 concrete structural assemblies.
- Implemented an Excel procurement tool, reducing production delays by 20% weekly across field and vendor teams.
- Authored 4 technical & customer-facing reports supporting quality documentation & life cycle verification.

**Experimental Aerodynamics Test Engineer |** *Wind Tunnel Undergraduate Research Group, University of Florida*
Gainesville, FL | August **2026** – Present

- Fabricated and assembled a Unistrut test rig (10 members cut to 20 ± 0.1 in.) for a 4-axis model positioning system beneath a subsonic wind tunnel test section.
- Performing flow characterization and instrument calibration at tunnel speeds up to 20 m/s to support model flight testing.
- Root-caused & resolved a fit interference with a laser-cut test section opening & insert, providing ± 180º yaw clearance.

**Fixed Wing UAS Design & Build |** *Independent Project*
Gainesville, FL | July **2026** – Present

- Building a 5-ft wingspan, ArduPilot GPS-guided UAS, integrating airframe, avionics, electrical, and propulsion subsystems.
- Reduced airframe mass 40% to 1.5 kg versus previous iteration by modifying material choices across foam & PLA subsystems.
- 3D printing a servo-actuated hemispherical camera gimbal with closed loop pointing control from 10 parts.
- Performed data driven stability analysis through Flow5 & ANSYS to inform airframe configuration and lift generation.
- Developed a structured flight test plan across a 16 week design-build-test cycle with root cause analysis.

## Education

**University of Florida,** Wertheim College of Engineering **Expected May 2028**

*Bachelor of Science in Aerospace Engineering* GPA: 3.6

**Relevant Coursework:** *Aerodynamics, Thermodynamics, Design & Manufacturing, Mechanics of Materials, Dynamics*

## Skills

- AI Prompting, Mechanical Design, Root Cause Analysis, SolidWorks/Fusion360 (Transferable to Creo), MATLAB, ANSYS, Rapid Prototyping, ASME Y14.5 GD&T, Design for Manufacturability, Microsoft Office

## Certifications

- CSWA Certified, *SolidWorks* Gainesville, FL | 2025 - Present
- Part 107 Remote Pilot License, *West Orange High School* Winter Garden, FL | 2023 - Present
- Part 61 Private Pilot License, *Orlando Flight School,* ***55 hours*** Orlando, FL | 2026 - Present